MILITARY SPECIFICATION

ANTENNA AS-1735/SRC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. This specification covers one type of ultra high frequency (UHF) antenna, the AS-1735/SRC, a vertically polarized, four-element array providing omnidirectional azimuth coverage over a frequency range from 225 to 400 megaHertz (MHz) when mounted around a central supporting mast ranging from 101.6 mm (four inches) to 381.0 mm (15 inches) in diameter.

2.0 APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-16400	Electronic Interior Communication And Navigation Equipment, Naval Ship And Shore, General Specifications For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions

DRAWINGS

NAVELEX

28687-0109350	AS-1735/SRC UHF Antenna Fabrication Plans

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-5043), Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1706, 222 South Riverside Plaza, Chicago, IL 60606.)

3.0 REQUIREMENTS

3.1 General. The UHF antenna shall be in accordance with the requirements of MIL-E-16400, except as otherwise specified herein.

3.2 First article. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.3 Design and construction. The UHF antenna shall be a single, interconnected unit consisting of four individual dipole elements mounted at 90 degree intervals in a horizontal plane about a central support mast. Internal interconnecting radio frequency (RF) feed lines and impedance matching sections shall permit simultaneous excitation of all four dipole elements from a single RF input connector. The antenna shall be adaptable for installation around central support masts ranging between 101.6 mm (four inches) to 304.8 mm (12 inches) in diameter and shall have adapter assemblies capable of allowing installations on mast sizes of 304.8 mm (12 inches) to 381.0 mm (15 inches) in diameter.

3.4 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the requirements of MIL-E-16400, except as specified in NAVELEX Drawing 28687-0109350.

3.5 Environmental requirements. Environmental service requirements shall be in accordance with the requirements of MIL-E-16400, except as specified in 3.5.1 through 3.5.8.

3.5.1 Temperature. The equipment shall conform to the Range 2 temperature requirements of MIL-E-16400.

3.5.2 Pressurization. The UHF antenna shall be operated unpressurized.

3.5.3 Salt fog (spray). The equipment shall be capable of withstanding the effects of a humid salt atmosphere, as specified in MIL-E-16400 for exposed equipment.

3.5.4 Wind velocity. The antenna shall operate normally in winds having a relative velocity of 75 miles per hour (MPH) and shall withstand without damage winds having a relative velocity of 120 MPH.

3.5.5 Hydrostatic pressure The hydrostatic pressure requirement of MIL-E-16400 is not applicable.

3.5.6 Underwater explosion. The underwater explosion requirement of MIL-E-16400 is not applicable.

3.5.7 Nuclear air blast. The nuclear air blast requirement of MIL-E-16400 is not applicable.

3.5.8 Gun blast. The gun blast requirement of MIL-E-16400 is not applicable.

3.5.9 Sunshine. The sunshine requirement of MIL-E-16400 is not applicable.

3.6 Mechanical requirements.

3.6.1 General. The AS-1735/SRC antenna shall be fabricated in accordance with NAVELEX Drawing 28687-0109350.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

3.6.2 Dimensions and weight. The AS-1735/SRC antenna shall have a maximum height of 711.2 mm (28 inches), and a maximum diameter not to exceed 762.0 mm (30 inches) for a 101.6 mm (four inches) to 304.8 mm (12 inches) central mast mounting. The maximum diameter shall not exceed 1090.6 mm (39 inches) with adapters for a 304.8 mm (12 inches) to 381.0 mm (15 inches) central mast mounting. The weight of the complete antenna shall not exceed 22.7 kilograms (50 pounds).

3.7 Electrical requirements.

3.7.1 Radiation pattern. The maximum radiation pattern will occur at zero elevation angle over the 225 to 400 MHz frequency band. The horizontal pattern of this antenna will be omni-directional to within ± 2 decibels (dB) over the entire 225 to 400 MHz frequency range when mounted around a central support mast 101.6 mm (four inches) to 381.0 mm (15 inches) in diameter.

3.7.2 Frequency range. The antenna and feed-system shall operate over the frequency range of 225 to 400 MHz.

3.7.3 Polarization. The antenna shall transmit or receive vertically polarized RF energy.

3.7.4 Gain. The gain of the antenna shall be approximately 2.1 dB above an isotropic level.

3.7.5 Power rating (RF). The antenna shall be capable of handling 2000 watts peak-envelope-power (PEP) RF power over the frequency range.

3.7.6 Voltage standing wave ratio. The impedance at the input to the antenna shall match the 50-Ohm impedance of the input coaxial cable so that the voltage standing wave ratio (VSWR) does not exceed 1.8:1 over the entire 225 to 400 MHz range when mounted.

3.7.7 Radio frequency connectors. The RF input to the antenna shall be by means of a connector capable of mating with Connector UG-154B/U.

3.8 Maintenance and repair. The antenna shall be so constructed that maintenance and repair can be made with a minimum of disturbance to other parts. The antenna shall be readily demountable and interchangeable to provide ease of servicing, maintenance, and replacement.

3.9 Workmanship. Workmanship shall be as specified in MIL-E-16400.

4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and b) Government product inspection to measure quality of product to be offered in acceptance. Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product criteria has been demonstrated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

4.1.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The methods of inspection fall within the following classifications:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)

1. Production inspection (4.4.1)
2. Production control inspection (4.4.2)

c. Environmental inspection (4.4.3)
d. Inspection of preparation for delivery (4.6)

4.3 First article inspection. Unless otherwise specified (see 6.3) one unit shall be required for first article inspection. First article inspection shall consist of all examinations and tests necessary to verify compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Surface examination	3.4, 3.6	4.5.1
Performance test		4.5.2
Temperature	3.5.1	4.5.3
Humidity	3.5.1	4.5.4
Salt fog (spray)	3.5	4.5.5
Fungus	3.5	4.5.6
Wind velocity	3.5.4	4.5.7
Icing	3.5	4.5.8
Shock	3.5	4.5.9
Vibration	3.5	4.5.10

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be in accordance with the production inspection requirements of MIL-E-16400.

4.4.2 Production control inspection. Production control inspection shall be conducted on a random sampling of units that have passed production inspection. Sampling shall be in accordance with the requirements of MIL-STD-105, Inspection Level S3, with an acceptable quality level (AQL) of 4.0. Tests shall include the temperature, humidity, and vibration tests of TABLE I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

4.4.3 Environmental inspection. Environmental inspection shall be conducted on one out of every 100 units, or fraction thereof, produced. Tests shall include the tests of TABLE I not required by 4.4.2

4.4.4 Reinspection of conforming production control and environmental test units. Unless otherwise specified, sample units which have been subjected to and passed the production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass the production inspection after repair of all visible damage.

4.5 Test methods.

4.5.1 Surface examination. Surface examination shall be in accordance with the requirements of MIL-E-16400.

4.5.2 Performance test. Both horizontal and vertical patterns, gain measurements, and power handling capability shall be taken at both ends and mid-band frequencies in accordance with contractor test procedures approved by the procuring activity.

4.5.2.1 Voltage standing wave ratio (VSWR). VSWR measurements shall be taken for five frequencies across the band in accordance with contractor test procedures approved by the procuring activity.

4.5.3 Temperatures. Temperature test shall be in accordance with the requirements of MIL-E-16400 for Range 2 temperatures.

4.5.4 Humidity. Humidity tests shall be in accordance with the requirements of MIL-E-16400.

4.5.5 Salt fog (spray). Salt fog (spray) tests shall be in accordance with the requirements of MIL-E-16400.

4.5.6 Fungus. Fungus tests shall be in accordance with the requirements of MIL-E-16400.

4.5.7 Wind velocity. Conformance to wind velocity requirements shall be shown by static load test or engineering analysis.

4.5.8 Icing. Conformance to icing requirements shall be shown by static load test or engineering analysis.

4.5.9 Shock. Shock test shall be in accordance with the requirements of MIL-E-16400.

4.5.10 Vibration. Vibration test shall be in accordance with the requirements of MIL-E-16400.

4.6 Inspection of preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

5.0 PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

6.0 NOTES

6.1 Intended use. The AS-1735/SRC ultra high frequency (UHF) antenna is intended for use on board ship to provide omnidirectional azimuth coverage over a frequency range from 225 to 400 megaHertz (MHz).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples if other than specified in 4.3
c. Preservation and packaging, packing, and marking requirements if other than specified in 5.1 (see 5.2)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Changes from previous issue. The extent of changes (deletions, additions, and so forth) precludes the annotation of the individual changes from the previous issue of this document.

Preparing activity:
NAVY-EC
(Project 5985-N466)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-A-24233A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

[ ] YES [ ] NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

M

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:21Z
Check the source to verify that this is the current version before use.